

Quiz (Transactions and Locks II)

Instructions: For each problem, you must perform all of the following tasks correctly in order to get credit.

1. Circle all SQL commands that block.
2. Write an X through all commands that error.
3. If a command causes a deadlock, write DEADLOCK and stop the problem.
4. Write the output of each SELECT statement.

You are allowed to use a computer and access the internet. (I strongly encourage you to refer to the postgres documentation.) BUT YOU ARE NOT ALLOWED TO CONNECT TO POSTGRES AND RUN POSTGRES COMMANDS.

Problem 1:

Session 1	Session 2
1 CREATE TABLE t (a INT UNIQUE);	1
2 CREATE TABLE u (a INT);	2
3 INSERT INTO t VALUES (1);	3
4 INSERT INTO t VALUES (2);	4
5 INSERT INTO u VALUES (1);	5
6 INSERT INTO u VALUES (2);	6
7 BEGIN;	7
8 LOCK TABLE t IN SHARE MODE;	8
9	9 BEGIN;
10	10 LOCK TABLE u IN
11	11 SHARE ROW EXCLUSIVE MODE;
12 LOCK TABLE u IN SHARE MODE;	12
13	13 LOCK TABLE t IN
14	14 SHARE UPDATE EXCLUSIVE MODE;
15 COMMIT;	15
	16 ABORT;

Problem 2:

Session 1	Session 2
1 CREATE TABLE t (	1
2 a INT PRIMARY KEY	2
3 DEFERRABLE INITIALLY DEFERRED	3
4);	4
5 BEGIN ISOLATION LEVEL	5
6 REPEATABLE READ;	6
7 INSERT INTO t VALUES (5);	7
8	8 BEGIN;
9	9 INSERT INTO t VALUES (6);
10 INSERT INTO t VALUES (6);	10
11 SELECT count(*) FROM t;	11
12	12 INSERT INTO t VALUES (5);
13	13 ABORT;
14 COMMIT;	

Session 1

```
1 CREATE TABLE t(a INT UNIQUE);
2 CREATE TABLE u(b INT REFERENCES t(a));
3 BEGIN;
4
5
6 INSERT INTO t VALUES (9);
7
8 COMMIT;
```

Session 2

```
1
2
3
4 BEGIN ISOLATION LEVEL
5 REPEATABLE READ;
6
7 SELECT count(*) FROM u;
8
9 INSERT INTO u VALUES (9);
10 COMMIT;
```

Session 1

```

1 CREATE TABLE t (a INT PRIMARY KEY);
2
3 BEGIN;
4 INSERT INTO t VALUES (1);
5 INSERT INTO t VALUES (2);
6 INSERT INTO u VALUES (2);
7
8
9
10 LOCK TABLE u IN ROW EXCLUSIVE MODE;
11 SELECT count(*) FROM t;
12
13
14 INSERT INTO t VALUES (3);
15 COMMIT;
16
17
18
19 SELECT count(*) FROM t;
```

Session 2

```

1
2 CREATE TABLE u (b INT NOT NULL);
3
4
5
6
7 BEGIN ISOLATION LEVEL
8 REPEATABLE READ;
9 INSERT INTO t VALUES (2);
10
11
12 INSERT INTO u VALUES (1);
13 SELECT count(*) FROM u;
14
15
16 UPDATE u SET b=5;
17 COMMIT;
18 SELECT count(*) FROM t;
```